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11 January 1983
SUPERSEDING
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15 November 1967

MILITARY SPECIFICATION

MUFFLERS, EXHAUST, INTERNAL COMBUSTION

ENGINE, GENERAL PURPOSE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers exhaust mufflers for general purpose application on internal combustion engines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2990

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MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-14105	- Paint, Heat-Resisting (For Steel Surfaces).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MIL-STD-1474	- Noise Level for Army Materiel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 176 - Stainless and Heat-Resisting, Chromium-Nickel Steel Plate, Sheet and Strip.
A 463 - Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The muffler shall be a special duct or pipe that impedes the transmission of sound while permitting the flow of exhaust gases from internal combustion engines.

3.2 Preproduction model. The contractor shall furnish a muffler for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce mufflers that comply with the requirements of this specification (see 6.3). Examination and test shall be those specified herein.

3.3 Material. The mufflers shall be constructed of materials which meet the specification of ASTM A176, UNS 540900 (iron-chromium stainless steel type 409), or ASTM A463 aluminum-coated steel. When tested as specified in 4.5.2.4, the loss in weight of material shall not be more than 8 grams per square foot of surface area.

3.4 Noise level. Noise levels shall not exceed the criteria stated in MIL-STD-1474. Noise levels for equipment not listed in MIL-STD-1474 shall be less than 85 dB(A) at the operator's position.

3.5 Back pressure. The muffler shall create a back pressure of not more than 8 inches of water for engines up to and including 10 horsepower, and not more than 20 inches of water for engines over 10 horsepower.

3.6 Resistance to air leakage. The muffler shall not leak more than 3-1/2 cfm of air when subjected to internal air pressure of 4.5 psi pressure.

3.7 Welding. When specified (see 6.2) the end caps and shell shall be welded in accordance with MIL-STD-1261.

3.8 Interchangeability. All mufflers having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Identification marking. The mufflers shall be identified in accordance with MIL-STD-130. In addition, the muffler inlet shall be clearly indicated.

3.10 Treatment and painting. When specified (see 6.2) the muffler shall be cleaned and treated in accordance with MIL-T-704 and painted in accordance with MIL-P-14105, color optional.

3.11 Workmanship. The mufflers shall be free from sharp bends and edges, defective joints, and other defects which would affect the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified,

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the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The muffler shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The mufflers shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.1 through 4.5.2.4. AQL shall be 6.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The mufflers shall be examined for the following defects:

101. Muffler assembly incorrect.
102. Treatment and painting not as specified (when specified).
103. Identification marking illegible, or not as specified.
104. Inlet not clearly indicated.
105. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Noise level. The muffler shall be tested as specified in SAE J366b, J1174, or J1207, as applicable. At least two readings shall be made to verify the results. A difference of more than 2 dB between the two readings at the same

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frequency indicates that the readings are inconclusive and shall be repeated. A noise level at or above 85db(A) at the operator's position, or a noise level above the level specified for equipment in MIL-STD-1474 shall constitute failure of this test.

4.5.2.2 Back pressure. The back pressure test may be run concurrently with the noise level test. The exhaust back pressure shall be measured by a manometer. The point of measurement shall be no more than 2 inches beyond the exhaust manifold outlet flange at rated load. The measurement shall be taken on a neutral flow plane at least one pipe diameter away from any changes in flow direction. Not less than 2 tests shall be made and the results averaged. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.3 Leakage. The inlet and outlet of the muffler shall be sealed. A means shall be provided to permit pumping 4.5 psi of air into the muffler. Evidence of leakage in excess of 3-1/2 cfm shall constitute failure of this test.

4.5.2.4 Corrosion resistance. The corrosion resistance test shall be performed on samples of material which are identical with the material used in the mufflers.

1. Two test panels, each 2- by 3-inches shall be used for this test. A 3/16-inch hole shall be punched near one 2-inch side of each panel. The panels shall be suspended by a glass rod through the hole, and the panels shall be separated by a glass spacer. The panels shall be free of corrosion products at the start of the test, and shall not be handled with the bare hands throughout the test.
2. The panels shall be degreased with acetone.
3. The panels shall be air dried at room temperature for not less than 15 minutes.
4. The panels shall be weighed to the nearest milligram.
5. The panels shall be dipped for not less than one minute in 1500 ml of aqueous solution consisting of 0.05N hydrobromic acid and 0.10N sulfuric acid at 200° F. The aqueous solution shall be contained in a 3000 ml Pyrex or Vycor Griffin beaker and shall completely cover the test panels.
6. The panels shall be raised and suspended one inch above the hot solution for 15 minutes. The beaker shall be covered to minimize loss of vapor.
7. Steps 5 and 6 shall be repeated and the panels placed in a muffle furnace at 1100° F for one hour.
8. The panels shall be removed from the furnace and allowed to cool to room temperature.
9. The panels shall be brushed with a brass brush to remove loose corrosion products.
10. Steps 5 through 9 shall be repeated three more times.
11. Steps 2, 3, and 4 shall be repeated.
12. A difference in weight loss between the two test panels greater than 3 grams per square foot indicates that the test is inconclusive and should be repeated.
13. Weight loss shall be multiplied by 12 to convert weight loss to grams per square foot.
14. A weight loss exceeding 8 grams per square foot shall constitute failure of this test.

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Notes:

a. The aqueous solution shall be changed every four cycles.
b. Duplicates of active metals (zinc, iron) shall not be run in the same corrosive bath.
c. Iron and zinc may be run together if desired.
d. A minimum specific ratio of 25 ml of the corrosive medium per square inch of specimen surface should be used.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Boxes for preservation or packing not as specified for level A or B.
107. Contents not immobilized as specified for level A.
108. Box not closed and sealed as specified for level A.
109. Quantities packed together exceed the size or weight limitation of the box for level A or B.
110. Preservation and packing not in accordance with the referenced document as specified for commercial.
111. Marking missing, illegible, incorrect, or incomplete for level A, B or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Each muffler shall be placed in a close-fitting, end-opening box, conforming to PPP-B-636, Class Weather-Resistant, grade as applicable to the weight of contents, style FOL. The contents shall be immobilized within the box in a manner to prevent movement or damage. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Level B. Each muffler shall be placed in a close-fitting, end-opening box, conforming to PPP-B-636, class domestic, grade as applicable to the weight of contents. The contents shall be immobilized within the box in a manner to prevent movement or damage. Box closure and sealing shall be as specified for method II, in accordance with the appendix to the box specification.

5.1.3 Commercial. Each muffler shall be preserved in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be level A, level B or Commercial as specified (see 6.2)

5.2.1 Level A. Mufflers of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Mufflers of like description, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional or boxes conforming to PPP-B-640, Class 2, style optional, may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Mufflers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The mufflers are intended to reduce the exhaust noise level of internal combustion engines. The recommended maximum sound levels, shown in MIL-STD-1474, Table 4, may be used as a guide in the selection of mufflers for use in various areas.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Time frame required for submission of the preproduction model (see 3.2).
c. Maximum noise level required if other than as specified in 3.4.
d. When the end caps and shell shall be welded (see 3.7).
e. When the muffler shall be painted (see 3.10).
f. Degree of preservation and degree of packing required (see 5.1, 5.1.2, and 5.1.3).

6.3 Preproduction model. Any changes or deviations of production mufflers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish mufflers conforming to this specification.

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
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User activity:
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER (*Code)* – Optional
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